

Group 2: Assignment 4

Movie Rating Classification and Movie Recommendation using RDR

Idea

- Assignment is to implement RDR interactive classification
 - *RDR learner* for objective standard
 - *RDR inter* for each team member's rules
- Applicable in real-world
- Real-world dataset must exist
- Can we do a recommendation system based on RDR?
 - Examples: laptop recommendation, university recommendation, career recommendation, ...
- Which application is most aligned with these characteristics?

Dataset

- Kaggle: <https://www.kaggle.com/datasets/tmdb/tmdb-movie-metadata>
- tmdb_5000_credits.csv (dump)
- tmdb_5000_movies.csv (choose)
 - 5000 rows
 - 20 cols

Process

- Copy and adapt relevant files from Sayar's repo
 - RDR files (no change)
 - inter/scrdr_interactive.py
 - classi/scrdr_learn.py
 - Bash files (adapt)
 - inter/run_inter_students.sh
 - classi/run_scrdr.sh
- Preprocess dataset (tmdb_5000_movies.csv)
- Shuffle and sample 50 rows
 - Setting RDR rules for 5000 rows is infeasible!

Movie Rating Classification

- Target = *rating_class* (Poor, Average, Good, Excellent)
- Exclude = *title, genres, original_language, vote_average*
 - Target is derived from *vote_average* → including = data leakage
- Build rules
 - Set rules based on data
 - But data is not always intuitive!
- Track logs, rules

Movie Rating Classification: Result

```
[Row 50/50] ERROR Found!
{'budget': 28934604.0, 'popularity': 9.642041, 'revenue': 14796236.0, 'runtime': 88.0, 'vote_count': 167, 'release_year': 2008, 'rating_class': 'Average'}

[KNOWLEDGE ACQUISITION]
System predicted: 'Good' but Actual is: 'Average'
Enter feature name (or 'exit'): runtime
Enter operator (==, <, >): <
Enter threshold value: 90

--- FINAL PERFORMANCE SUMMARY ---
      precision    recall  f1-score   support

   Average         1.00      0.75      0.86         4
  Excellent         1.00      1.00      1.00         5
     Good          0.98      0.98      0.98        41
    Unknown         0.00      0.00      0.00         0

 accuracy          0.96         50
  macro avg         0.74      0.68      0.71         50
 weighted avg         0.98      0.96      0.97         50

Loaded existing model: ./rules/inter_movies_rules.json
--- Running in TEST mode ---

--- FINAL PERFORMANCE SUMMARY ---
      precision    recall  f1-score   support

   Average         1.00      0.50      0.67         4
  Excellent         1.00      1.00      1.00         5
     Good          0.95      1.00      0.98        41

 accuracy          0.96         50
  macro avg         0.98      0.83      0.88         50
 weighted avg         0.96      0.96      0.95         50
```

Movie Recommendation

- Title cannot be the target
 - Predicting movie title is not what we want
- Need a new variable
 - Copy a 50-sample dataset
 - Add coarse decision label
- Target = *recommend* (Recommend, Not)
- Exclude = *budget, popularity, revenue, vote_count, rating_class*
 - Include = *genres, runtime, vote_average, release_year*
- Treating *genres* needs a modified ``scrdr_interactive``
- Build rules
- Track logs, rules

Movie Recommendation: Result

```
--- MATCHED TITLES (predicted conclusion in in_set) ---
in_set: ['Recommend']
Split Second
Harry Potter and the Half-Blood Prince
Taxi Driver
End of Days
My Own Private Idaho
(500) Days of Summer
The Artist
Kill Bill: Vol. 1
Once Upon a Time in the West
Rocky
The Celebration
[count] 11
Loaded existing model: ./rules/modified_inter_movies_rules.json
--- Running in TEST mode (target=recommend, title_col=title, genre_col=genres, in_set=['Recommend']) ---

--- FINAL PERFORMANCE SUMMARY (coarse labels) ---
      precision    recall  f1-score   support

     Not         0.92      0.92      0.92        38
    Recommend     0.82      0.75      0.78        12
     Unknown     0.00      0.00      0.00         0

 accuracy         0.88         50
 macro avg       0.58       0.56       0.57        50
weighted avg       0.90       0.88       0.89        50

--- MATCHED TITLES (predicted conclusion in in_set) ---
in_set: ['Recommend']
Split Second
Harry Potter and the Half-Blood Prince
Taxi Driver
End of Days
My Own Private Idaho
(500) Days of Summer
The Artist
Kill Bill: Vol. 1
Once Upon a Time in the West
Rocky
The Celebration
[count] 11
(lu_ai_engineering_class_env) lawun330@ROG-Strix-Hero-II:~/Desktop/AIE-F/assignment-submission/class-10/group-2$
```

Conclusions / Findings

- RDR may not scale well with data growth
- Interactive RDR takes too much time w.r.t automated RDR
- Data is not always intuitive
- Recommendations can accidentally go against user preferences (e.g., presenting the wrong genres)
 - Preferred - Action, Adventure, Thriller
 - 2500000.0,"['Drama', 'Romance']",en,10.363642,121566957.0,104.0,**My Own Private Idaho**,7.1,212,1991,Good,Recommend

Thank you!!

Any Questions?